

VARUN SHARMA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with over 2+ years of experience at TCS in backend development and cloud infrastructure. Specialized in building scalable Node.js microservices and managing **AWS** environments. Skilled in improving data retrieval using **Elasticsearch** and **OpenSearch**, with a focus on maintaining high system reliability and 99.9% uptime for mission-critical healthcare applications.

TECHNICAL SKILLS

- **Languages:** JavaScript, Java, Prompting
- **Backend:** Node.js, Express.js, REST API Design, Microservices
- **Cloud & DevOps:** AWS (EC2, ECS, Lambda, S3, CloudFormation, CloudWatch), Docker, Kubernetes
- **Search & Databases:** Elasticsearch (Query DSL, Custom Analyzers), OpenSearch (Vector Search), Redis
- **Tools & Methods:** Git, System Design, Data Structures (DSA), Agile/Scrum, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Software Developer

Dec 2023 – Present

Tata Consultancy Services (TCS)

- Developed and deployed Node.js microservices and REST APIs for high-traffic healthcare systems, improving overall system modularity and speed.
- Automated cloud setup using **AWS CloudFormation**, reducing manual errors and ensuring consistent deployments across testing and production environments.
- Optimized Elasticsearch clusters for a **500k+ record** dataset, reducing search latency by **40%** using custom analyzers and NLP query tuning.
- Set up real-time monitoring and alerting via **AWS CloudWatch**, maintaining **99.9% uptime** for patient-critical data services.
- Collaborated in an Agile team to convert business requirements into technical solutions, delivering key product features on schedule.

Full Stack Developer Intern

Jan 2023 – June 2023

MRIIC

- Built full-stack features using the **MERN stack and Django** for internal tools, improving the efficiency of administrative tasks.
- Integrated secure RESTful endpoints to sync data between React frontends and cloud-hosted databases.
- Improved user interfaces with Bootstrap and JavaScript, leading to better cross-device usability and user engagement.

PROJECTS

Vector-Based Search Engine | *NLP, OpenSearch, Node.js*

[Link](#)

- Created a semantic search system using **all-mpnet-base-v2** embeddings to provide accurate search results based on user intent rather than just keywords.
- Built a Node.js backend to convert queries into **768-dimensional vectors** for real-time similarity matching in **OpenSearch**.
- Designed a custom vector index schema that improved search precision by 35% for complex, conceptual product searches.

EDUCATION

Manav Rachna International University

2019 – 2023

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Faridabad, Haryana

CERTIFICATIONS & ACTIVITIES

- **Technical Training:** **Kubernetes** and **Docker** certifications via KodeKloud.
- **Design Certification:** **Figma UI/UX Design Training** for building professional prototypes.
- **Mentorship:** Guided junior developers on **Django** and **Python** during the Samsung Hackathon.
- **Activities:** Represented the university in table tennis and performed as a guitarist in music events.